FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility

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STANDARD OPERATING PROCEDURE

Startup Procedure for Water Treatment Line 20

Department: Water Treatment

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Prepared by: Engineering / QA Department
Approved by: Plant Manager

Document Control Information

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Title	Startup Procedure for Water Treatment Line 20
Department	Water Treatment
Plant	FreshFlow Bottling Plant - Pune Production Facility
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1. Purpose

Produce water meeting the FreshFlow Beverages water quality specification (TDS <50 ppm, pH 6.5-7.5, microbiologically safe) for use in product production and CIP.

2. Scope

Water Treatment operators and Maintenance team.

3. Prerequisites and Requirements

3.1 Training Requirements

- RO system operation training
- Chemical dosing training

3.2 Tools and Materials Required

- Calibrated instruments as listed in the plant calibration schedule.
- Appropriate PPE as specified in Section 4.
- Completed permit-to-work (if required for maintenance activities).
- Access to relevant equipment manual and P&ID drawings.
- CMMS access to create and close work orders.

4. Health, Safety and Environmental Requirements

! WARNING

This SOP must not be carried out without appropriate PPE and, where required, a valid Permit to Work (SOP-SAF-006). If in doubt about safety, STOP and consult the Safety Officer (EMP009 / EMP017).

4.1 Identified Hazards

- High-pressure RO system (up to 12 bar) - pressure release risk
- UV radiation from UV sterilizer lamps - eye and skin damage risk
- Chemical dosing agents (chlorine, antiscalant, acid) - chemical hazard

4.2 Emergency Response

In case of any emergency during this procedure: Press the nearest Emergency Stop, alert personnel in the vicinity, call the Safety Officer and Shift Supervisor. Refer to the Emergency Response Plan (SOP-SAF-004) and site evacuation procedure.

5. Detailed Procedure

5.1 Pre-Task Verification

1. Confirm the relevant equipment is available and no conflicting activities are planned.
2. Check shift handover log for any outstanding issues with this equipment.
3. Verify all required materials, tools, and chemicals are available and correctly labelled.
4. Obtain all required permits (PTW, hot work, confined space as applicable).
5. Brief all involved personnel on this SOP before starting.

5.2 Main Procedure

1. Verify all LOTO devices removed. Confirm no maintenance work in progress on this equipment.
2. Check all fluid levels: oil, coolant, and product supply as applicable.
3. Ensure all guards and safety interlocks are fitted and functional - test before starting.
4. Confirm utility services available: compressed air 6-7 bar, steam 5.5-7 bar, chilled water 2°C +/- 1°C.
5. Navigate to the equipment tag on SCADA. Acknowledge start permissives.
6. Start the equipment using the approved sequence. Observe initial parameter readings for 5 minutes.
7. Verify all parameters (temperature, pressure, flow, current) are within normal range (see equipment manual).
8. Record startup time, initial parameter readings, and operator name in the shift log.